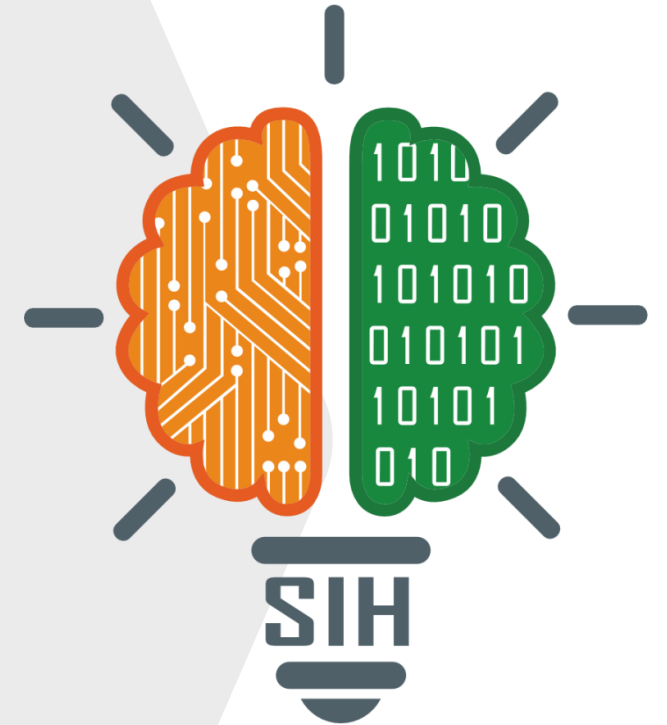


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** - SIH26104
- **Problem Statement Title** - AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- **Theme** - Blockchain & Cybersecurity
- **PS Category** - Software
- **Team ID** - NA
- **Team Name** – Power Rangers



VigilVoice - Voice Trust Gate

Detection buys evidence, prevention requires action

Proposed Solution

"Converts voice-risk evidence into an authorization decision — before the action completes."



The idea

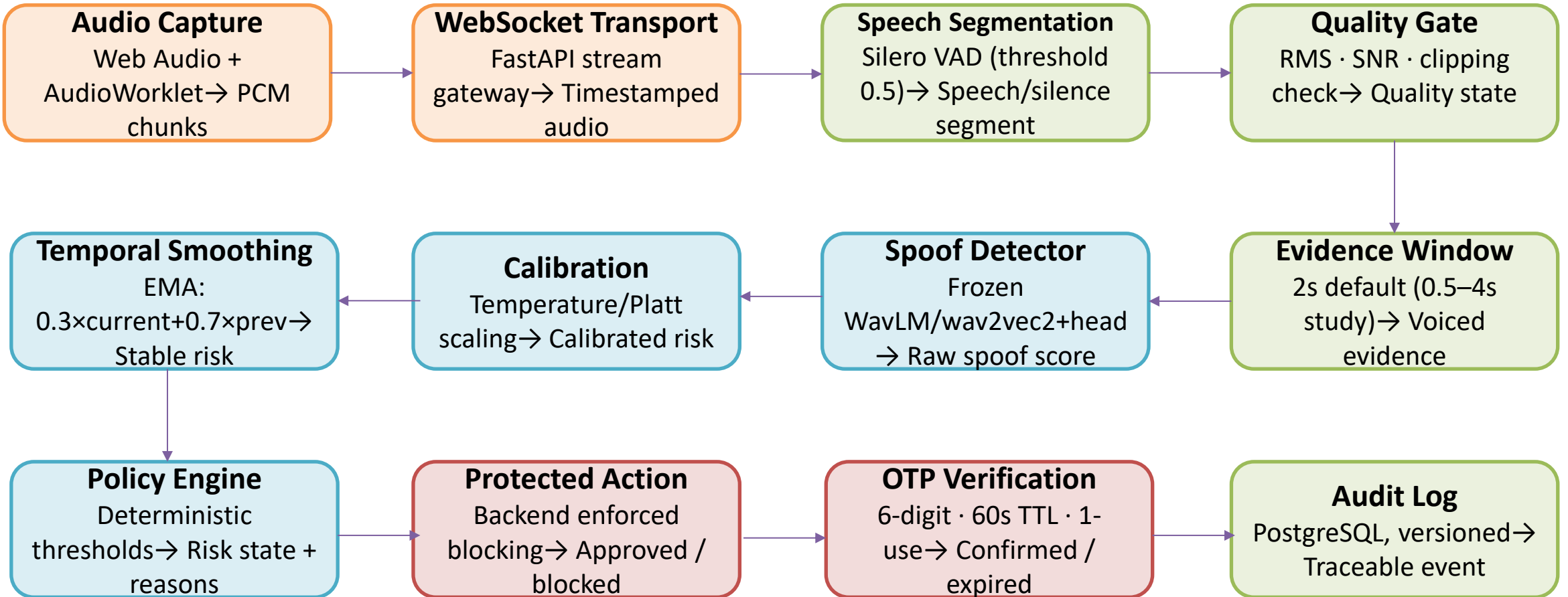
- Continuous voice analysis
- Dynamic risk scoring
- Step-up verification
- Action blocking + audit trail

Addresses the problem

- Detectors fail on unseen generators, codecs, replay
- Benchmark accuracy $\neq$ production safety
- Acts before the action completes, not just flags it

Uniqueness

- Prevention-first, not detection-only
- Uncertainty-aware, fails safely
- Indian-language evaluated (Hindi, English, Hinglish, regional)
- Fully auditable decisions



Existing open datasets IndicSynth, IndicVoices, ASVspoof 5, RTCFake, ReplayDF

Offline/local fallback

local model, clips and OTP, file-upload mode, zero dependency on internet APIs

Proven open components

Silero VAD, frozen WavLM/wav2vec2, FastAPI, PostgreSQL, Docker Compose

Modest hardware

8-core CPU / 16GB RAM minimum, GPU optional

36-hour buildable roadmap

Phase 0 (prep) → Phase 6 (rehearsal)

Lean MVP scope

one VAD + one spoof detector + policy engine, no multi-model ensemble

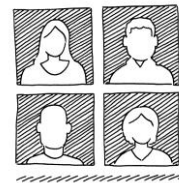
- **Earned-complexity roadmap** — TrustGraph adds branches (speaker verification, replay, OOD, few-shot) only after validation, not upfront
- **Fail-closed by design** — every failure mode (poor audio, stream loss, OTP expiry, detector crash) keeps the action unapproved
- **Privacy-by-default** — no raw audio storage, hashed OTP/tokens, auditable events (without overclaiming legal compliance)
- **Tested under real conditions**, not just benchmarks — language, channel, duration, unseen-generator, replay
- **Disciplined claims** — avoids overclaiming, builds long-term institutional trust
- **Low-cost to operate at scale** — GPU-optional, open-source stack

Who is this built for, and how does it change their world?

- **Banks & Fintech** Fund-transfer and other sensitive approvals get an authorization gate before they complete
- **Telecom / Call Channels** System is validated against real telephone conditions — codec loss, noise, RTC degradation
- **General Public** Genuine Hindi/Hinglish/regional-language callers face less friction, not more
- **Regulators & Auditors** Every decision carries reason codes, model & threshold version — but the team never claims legal-compliance certification

What good does it actually produce?

- **Social** Shields Indian-language speakers from voice-clone fraud; privacy-preserving by default (no raw audio stored)
- **Economic** Blocks fraudulent transfers before completion; fewer false alerts = less friction; GPU-optional stack = low deployment cost
- **Governance & Trust** Full audit trail with reason codes; fail-closed under any uncertainty



- IndicSynth — synthetic Indian-language attacks
- IndicVoices — genuine Indian-language speech
- ASVspoof 5 — anti-spoof benchmark
- RTCFake — codec/channel conditions
- ReplayDF — replay attacks

Core techniques:

- Silero VAD
- WavLM/wav2vec2 (frozen SSL)
- LFCC-LCNN (fallback)
- AASIST (future branch)

Metrics we measure:

1. Detection rate at fixed FPR.
2. First-alert latency.
3. False alerts per genuine call.